

groups and rings



We have learned many nice results

on the integers. To name a few,

1, the division algorithm

2, prime = irreducible

3, unique prime factorization.

All of them fall into a bigger and

more abstract patterns in the study

of groups and rings.

Def 1: A nonempty set G is a group if there is binary operation

$$\cdot : G \times G \rightarrow G \quad (a, b) \mapsto ab$$

such that 1) (associativity),

$$\forall a, b, c \in G. \quad a(bc) = (ab)c$$

2) (identity),

$$\exists e \in G. \text{ s.t. } \forall g \in G. \quad eg = g = ge$$

3) (inverse),

$$\forall g \in G. \quad \exists g^{-1} \in G \text{ s.t. } g^{-1}g = e = gg^{-1}$$

Sometimes we write $(G, e, \cdot)$ to indicate group G .

eg: $(\mathbb{Z}, 0, +)$ is a group

$(\mathbb{N}, 0, +)$ is not a group

$(\mathbb{Z}_m, 0, +)$ is a group

Def 2. Let G be a group. We

say G is abelian : if $ab = ba \quad \forall a, b \in G$

Def 3. A set R is a unital ring

there are two binary operations

$$+ : R \times R \rightarrow R \quad (a, b) \mapsto a + b$$

$$\cdot : R \times R \rightarrow R \quad (a, b) \mapsto ab$$

such that

(1) $(R, 0, +)$ is an abelian group

(2) $(R, 1, \cdot)$ is a monoid.

i.e. $\exists$ an element $0 \neq 1 \in R$

such that $\forall r \in R$, $r \cdot 1 = r = 1 \cdot r$

and $\forall a, b, c \in R$, $a(bc) = (ab)c$

(3) multiplication is distributive over addition

i.e. $a \cdot (b+c) = ab + ac$

$(a+b) \cdot c = ac + bc$

Sometimes we say $(R, +, \cdot)$ to be the ring

eg. $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Z}_m, +, \cdot)$ are rings

Def 4. A ring is commutative if

$$\forall a, b \in R \quad ab = ba.$$

Def 5. Let R be a commutative ring.

(1) A nonzero element u is a unit

$$\text{if } \exists^{0 \neq} v \in R \quad \text{s.t.} \quad uv = 1$$

Sometimes we also call a unit invertible

(2) A nonzero element a is

$$\text{a zero-divisor if } \exists^{0 \neq} b \in R,$$

$$\text{s.t.} \quad ab = 0$$

Def 5. let R be a commutative ring.

(1) R is an (integral) domain

if R contains no zero-divisors

(2) R is a field if

all nonzero elements are invertible

eg. $\mathbb{Z}$ is an integral domain

let $n > 1$. Then $\mathbb{Z}_n$ is not

an integral domain if n not prime

and $\mathbb{Z}_n$ is a field iff

n is prime.

eg. let $R = \mathbb{Z}[\sqrt{2}]$

$$= \left\{ a + b\sqrt{2} : a, b \in \mathbb{Z} \right\}$$

We find the units in R .

Suppose $a + b\sqrt{2}$ is a unit

Then $\exists c + d\sqrt{2} \in R$ s.t

$$(a + b\sqrt{2})(c + d\sqrt{2}) = 1$$

Expand and we get

$$(ac + 2bd) + (ad + bc)\sqrt{2} = 1$$

$$\text{so } ac + 2bd = 1$$

$$ad + bc = 0$$

Note that if we replace

b with $-b$. d with $-d$

The equations remain unchanged.

That is

$$(a + b\sqrt{2}) \mid (c + d\sqrt{2}) = 1$$

$$\text{iff } (a - b\sqrt{2}) \mid (c - d\sqrt{2}) = 1$$

i.e. $a + b\sqrt{2}$ is a unit

with inverse $c + d\sqrt{2}$

iff $a - b\sqrt{2}$ is a unit

with inverse $c - d\sqrt{2}$

In particular,

$$(a+b\sqrt{2})(c+d\sqrt{2})(a-b\sqrt{2})(c-d\sqrt{2}) = 1$$

$$(a^2-2b^2)(c^2-2d^2) = 1$$

$$\text{So } a^2-2b^2 = \pm 1$$

Conversely, for $a+b\sqrt{2}$ where

$$a^2-2b^2 = \pm 1,$$

We note $(a+b\sqrt{2})(a-b\sqrt{2}) = \pm 1$

So $a+b\sqrt{2}$ is a unit

Combining both results.

Prop 6: Let $R = \mathbb{Z}[\sqrt{2}]$.

Then $a+b\sqrt{2}$ is a unit

$$\text{iff } a^2 - 2b^2 = \pm 1$$

What are the zero-divisors?

Let $a+b\sqrt{2}$ where $a, b \in \mathbb{Z}$

be a zero divisor.

Then $\exists c+d\sqrt{2}$ where $c, d \in \mathbb{Z}$

$$\text{s.t. } (a+b\sqrt{2})(c+d\sqrt{2}) = 0$$

$$\text{i.e. } ac + 2bd = 0$$

$$ad + bc = 0$$

Replace b with $-b$

d with $-d$

does not affect the equation

that is

$$(a+b\sqrt{2})(c+d\sqrt{2}) = 0$$

$$\text{iff } (a-b\sqrt{2})(c-d\sqrt{2}) = 0$$

In particular,

$$(a+b\sqrt{2})(c+d\sqrt{2})(a-b\sqrt{2})(c-d\sqrt{2}) = 0$$

$$(a^2-2b^2)(c^2-2d^2) = 0$$

$$\text{So } a^2-2b^2=0 \quad \text{or} \quad c^2-2d^2=0$$

No integers a, b, c, d solve these equations.

Exercise.

1. Show $\mathbb{Q}[\sqrt{2}]$ is a field

2. $\mathbb{Z}[\sqrt{2}] \subseteq \mathbb{Q}[\sqrt{2}]$

3. subring of field is domain

4. conclude $\mathbb{Z}[\sqrt{2}]$ domain

$\mathbb{Z}[\sqrt{2}]$ is ED.

classify the primes?

